

16. Answer: C

Intensity decreases with the order of the fringe image, due to the diffraction intensity envelope and that the higher order fringes will travel longer distances to the screen compared to those of lower order.

Using $d \sin \theta = n\lambda$

$$d = \frac{n\lambda}{\sin \theta}$$

For 1st order fringe image, $d = \frac{\lambda}{\sin \theta_1}$.

For 2nd order fringe image, $d = \frac{2\lambda}{\sin \theta_2}$.

$$\frac{\lambda}{\sin \theta_1} = \frac{2\lambda}{\sin \theta_2} \Rightarrow \sin \theta_2 = 2 \sin \theta_1$$

$\therefore 2\theta_2 < \theta_1$, because the gradient of the sine graph decreases with increasing angle θ , for $0^\circ < \theta < 90^\circ$.